A026 Hidden-State Memory & Backreaction

Independent research sidecar v0.1.0 - DEVELOPMENT ONLY / NOT AN EULER OR SWIRL CLOCK CERTIFICATION

01. Executive scientific verdict

This sidecar checks a narrowly defined question: can hidden conservative modes create observational memory and return energy when only one mode is measured? A synthetic benchmark validates the diagnostic, but does not certify a three-dimensional incompressible Euler field or an intrinsic SST Swirl Clock.

02. Research question

At identical currently observed knot state, can the prior observed trajectory improve held-out predictions because omitted background vorticity modes contain physical state information? Is any inferred phase frequency dynamically qualified?

03. Existing-family and novelty audit

A026 tests phase-feedback delay and Kelvin transport; A049 tests effective transverse rigidity; A068/A069 address phase clocks. This sidecar adds a matched-current-versus-history prediction comparison. It is not a replacement for, or a new certified version of, these families.

04. Sources and provenance

Paper: Muruga, Ginot, Loos and Bechinger (2026), Nature Physics, doi:10.1038/s41567-026-03442-w. Closest project source: A026 v0.2.1 README and A049 v0.4.1 README. Every generated run records its input and code hashes; a new external provider must be independently qualified.

05. Physics scope and assumptions

3D incompressible inviscid Euler, no forcing, is the future target. Current executable proof is a two-mode linear, conservative Hamiltonian benchmark with SI-unit positions and a reference mass of 1 kg. No molecular viscosity, stochastic bath, Euler pressure solver or knot topology is represented in the executable benchmark.

06. State-space and observables

Observable current features are x(t), xdot(t). History features are x(t-delay), xdot(t-delay). Oracle adds hidden y(t), ydot(t). Prediction target is x(t+horizon). A separate future Euler adapter must export a preregistered geometric scalar q and its true derivative, with independent trial IDs.

07. Equations and dimensionality

Benchmark: xddot + omegaK^2*x + g*y=0; yddot + omegaB^2*y + g*x=0. Here [omega]=s^-1, [g]=s^-2. Energy E=m/2*(xdot^2+ydot^2+omegaK^2*x^2+omegaB^2*y^2+2*g*x*y) has units joule. Eliminating y yields a sinusoidal memory kernel, not a viscous constitutive law.

08. Null, alternative and preregistered criteria

H0: with coupling g=0, past features do not improve future prediction beyond current position and velocity; upper 95% bootstrap limit on relative root-mean-square error gain must stay below 0.05. H1 synthetic: with g=2 s^-2, lower 95% gain bound must exceed 0.20. Euler science gates require new thresholds registered before exposure.

09. Dataset and sampling

Each synthetic sample is one independently seeded complete trajectory, sampled at fixed observation time t=2.1 s, delay 0.65 s and horizon 0.75 s. Training and test are split by complete trajectory. Independent bootstrap resamples test trajectories. Target-only Gaussian measurement noise is included to avoid zero-error ratios.

10. Comparison method

Fit ordinary least-squares predictions using train-only standardization: (i) current state, (ii) current plus history, and (iii) full-state oracle. Measure out-of-sample RMSE and improvement 1-RMSE_history/RMSE_current. Bootstrap 400 times on independent test trajectory IDs.

11. Verification and negative controls

Check analytic energy invariance, decoupled null, coupled positive, duplicate trial rejection, non-finite rejection and minimum independent sample size. The synthetic positive cannot be used as Euler evidence. Future controls need shuffled history with matched current, stationary-history drift checks, and uncertainty under finite measurement resolution.

12. Numerical qualification

For a real 3D Euler conclusion: independent spatial, temporal, finite-core and domain convergence at common final physical time; divergence/momentum, energy/helicity/circulation residuals and knot isotopy checks. A filament regularizer and spectral Gaussian core are not interchangeable. These tests are NOT EXECUTED here.

13. Hidden-state test result

The data table and numerical decision ledger appear below. The result qualifies the algebraic numerical method only; it does not prove a background network restores a trefoil or sustains a clock.

Case	RMSE current (m)	RMSE history (m)	Gain 95% CI
case_coupled	0.002932	0.0001865	[0.9304, 0.9419]
case_decoupled	0.0001866	0.0001866	[-0.0, 0.0]
case_history_shuffled	0.002932	0.00296	[-0.0155, -0.0036]

G0: **PASS** - Exact Hamiltonian energy max relative error 9.088e-16

G1: **PASS** - Decoupled null: gain upper 95% < 0.05

G2: **PASS** - Coupled positive: gain lower 95% > 0.20

G2B: **PASS** - Randomly permuted history must not improve the independent holdout prediction

G3: **PASS** - Full-state oracle closes the hidden-mode information gap in the synthetic linear system

G4: **NOT_EXECUTED** - No 3D Euler model/provider or dynamic background core supplied

G5: **NOT_EXECUTED** - Spatial, time, finite-core and box convergence missing

G6: **NOT_EXECUTED** - Native Euler energy, circulation, helicity, topology not independently measured

G7: **NOT_EXECUTED** - No phase-locked intrinsic clock, no operational tau measurement

G8: **NOT_EXECUTED** - No independently executed scientific 2-of-3 backend parity

G9: **NOT_APPLICABLE** - No reveal targets; no secret labels or fitted SST constants

14. Swirl Clock test

Absolute time t_abs is the underlying evolution parameter. An operational phase clock requires a stable phase theta and measured Omega=dtheta/dt_abs. Only after demonstrable phase coherence, environmental perturbation controls and full converged dynamics may d tau/dt_abs=Omega/Omega_ref be evaluated. No clock law is inferred in this release.

15. Falsifiers, confounders and status

Critical confounders: incomplete observation, noisy current state, remeshing, nonstationary sampling, shared trajectories across splits, numerical stabilization, and unresolved Euler core. Physical statuses remain UNRESOLVED / NOT EXECUTED, even if all synthetic smoke gates PASS.

16. Reproduction, outputs and bibliography

Run run_all.cmd TEST followed by run_all.cmd BASIC. Run EULER only with source-audited NPZ independent trajectories. Output produces machine-readable gate ledgers, science report and SHA-256 archive. Citations: Muruga et al. (2026), DOI 10.1038/s41567-026-03442-w; Mori (1965), DOI 10.1143/PTP.33.423; Sagawa & Ueda (2012), DOI 10.1103/PhysRevE.85.021104.